An Over View Immunization Programme

Topic 6

Introduction

Immunization is one of the most cost-effective health investments

Vaccination played a major role in the battle on infectious diseases since 496 B.C

Edward Jenner, pioneered the concept of vaccines and developed world's first vaccine for smallpox

Last case of smallpox in Somalia in 1977 and eradicated globally in 1980

In 1974, 27th World Health Assembly recommended vaccination against six diseases: Tuberculosis, Diphtheria, Tetanus, Pertussis, Measles and Poliomyelitis under Expanded Programme on Immunization (EPI).

EPI was launched in India in 1978 (for 6 VPDs :BCG, OPV, Diphtheria, Pertusis, Tetanus , Typhoid-Urban area)

India’s Universal Immunization Programme (UIP)

One of the largest public health programs in the world

1948
BCG vaccine was among the first vaccines introduced in India in limited area

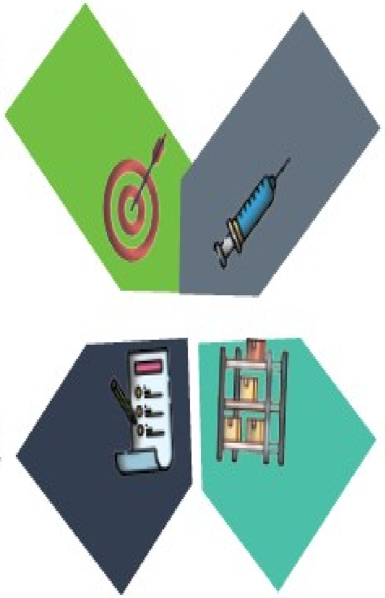
1962
BCG vaccination became part of the National Tuberculosis Control Programme

1978
Expanded Programme of Immunization (EPI) launched with BCG, DPT, OPV and Typhoid (Urban areas)

1985
Expanded as Universal Immunization Programme (UIP) with addition of measles and removal of Typhoid vaccine.
Focus on children less than 1 year of age.

Annual target
2.6 crore newborns
3.0 crore pregnant women

Vaccine against VPDs
11 nation wide
1 sub-nationally (JE)

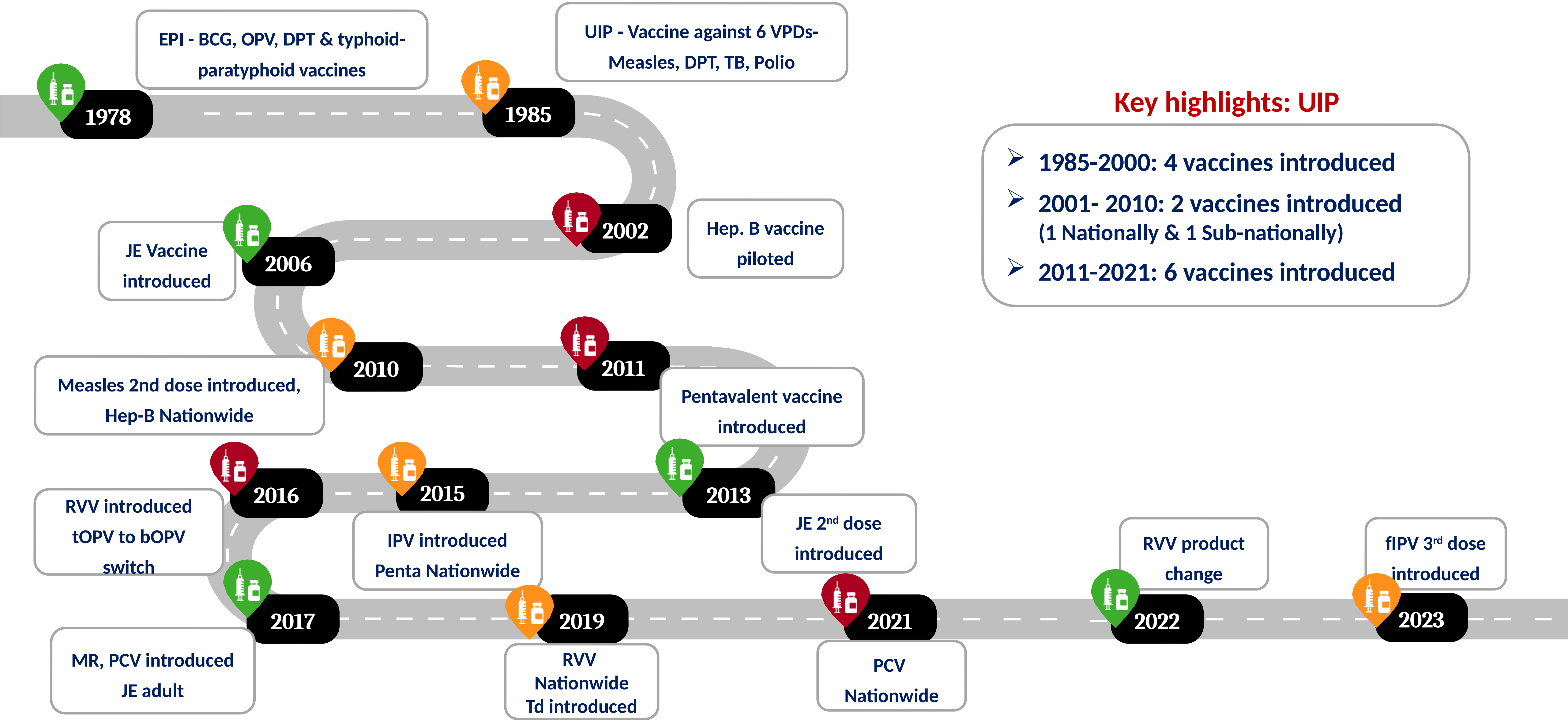


~1.3 crore sessions planned per year

~30,000 cold chain points for storage and distribution

All vaccines under UIP are manufactured in India
India has largest vaccine manufacturing capacity in the world

Roadmap of Introduction & Expansion of New Vaccines



Key highlights: UIP

- 1985-2000: 4 vaccines introduced
- 2001- 2010: 2 vaccines introduced (1 Nationally & 1 Sub-nationally)
- 2011-2021: 6 vaccines introduced

Universal Immunization Programme

(Scope and Scale in Odisha)

Annual target 2025-26

Newborns : 611249

Pregnant women : 726522

Vaccine against VPDs

12 Nos. of Vaccine with 13 Nos. of Vaccine Preventable Diseases including HPV

One of the largest Public Health Programmes

> 30000 sessions planned per year

~1293 cold chain points for storage and distribution of vaccines

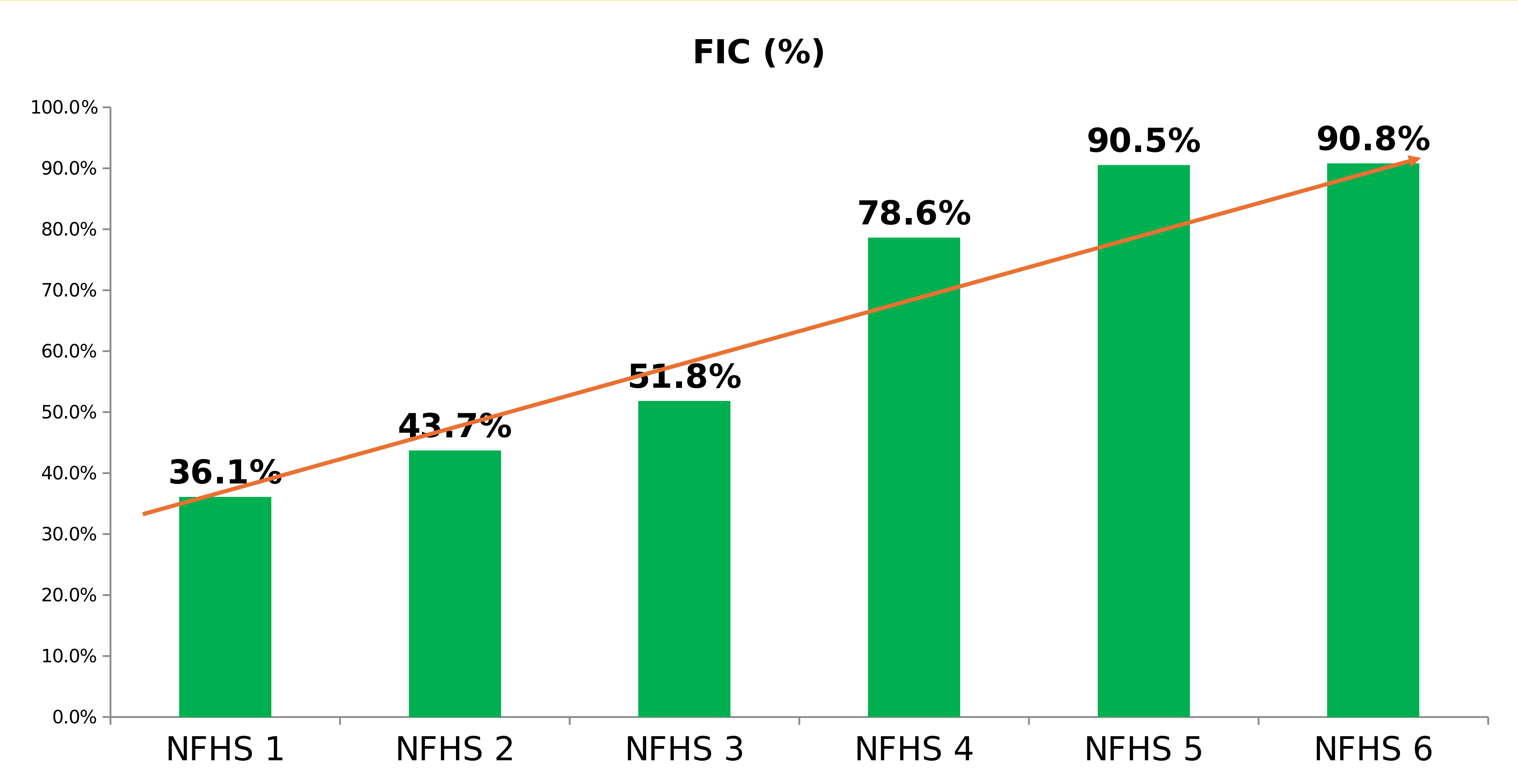
(NCCMIS)

Aim and objectives of UIP

Eligible beneficiaries should get all UIP vaccines, to be protected from life threatening Vaccine Preventable Diseases

- *To attain more than 90% full immunization coverage and sustain the gains*
- *To achieve MR elimination*
- *To maintain and sustain Polio free status and Maternal & Neonatal Tetanus elimination*
- *To sustain the robust VPD and AEFI surveillance*
- *To ensure efficient supply chain, cold chain and logistic system for UIP*
- *To provide regular supportive supervision to UIP*
- *To ensure accountability framework through regular task force meetings*
- *To expand digitization in UIP*
- *To introduce need based new vaccines*

Full Immunization trend (NFHS)



Concept of Full Immunization Coverage (FIC) Plus

Full Immunization Coverage (FIC) Plus –	*Child who has received BCG, 3 doses of OPV, 3 doses of pentavalent and 1 dose of MR by first year of age. This is the historical definition used for survey and evaluation purposes and hence has been continued. - Full Immunization coverage (FIC)*
FIC plus	*Child who has received all the vaccines given in the Universal Immunization Programme (UIP) by one year of age.*
Complete Immunization	Child who has received all the vaccines given in the Universal Immunization Programme (UIP) by two years of age
Zero-dose (operational & reporting purpose)	Child who has not received first dose of pentavalent vaccine by one year of age.

Immunization programme Component

1. VPDs and Management of outbreaks
2. VPD Surveillance .
3. Immunization Schedule.
4. Planning and Conducting Immunization
   - Preparation of Micro Plans
5. Vaccine and Other logistics Management
   - Alternate Vaccine Delivery
   - Management of vaccine, Immunization cards etc.
6. Cold Chain Management
7. Injection Safety .
8. Adverse event following immunization .
9. Records , Reports and Using data for action.
10. Monitoring & supportive Supervision.
11. Demand generation for improved coverage

VPDs

Viral diseases

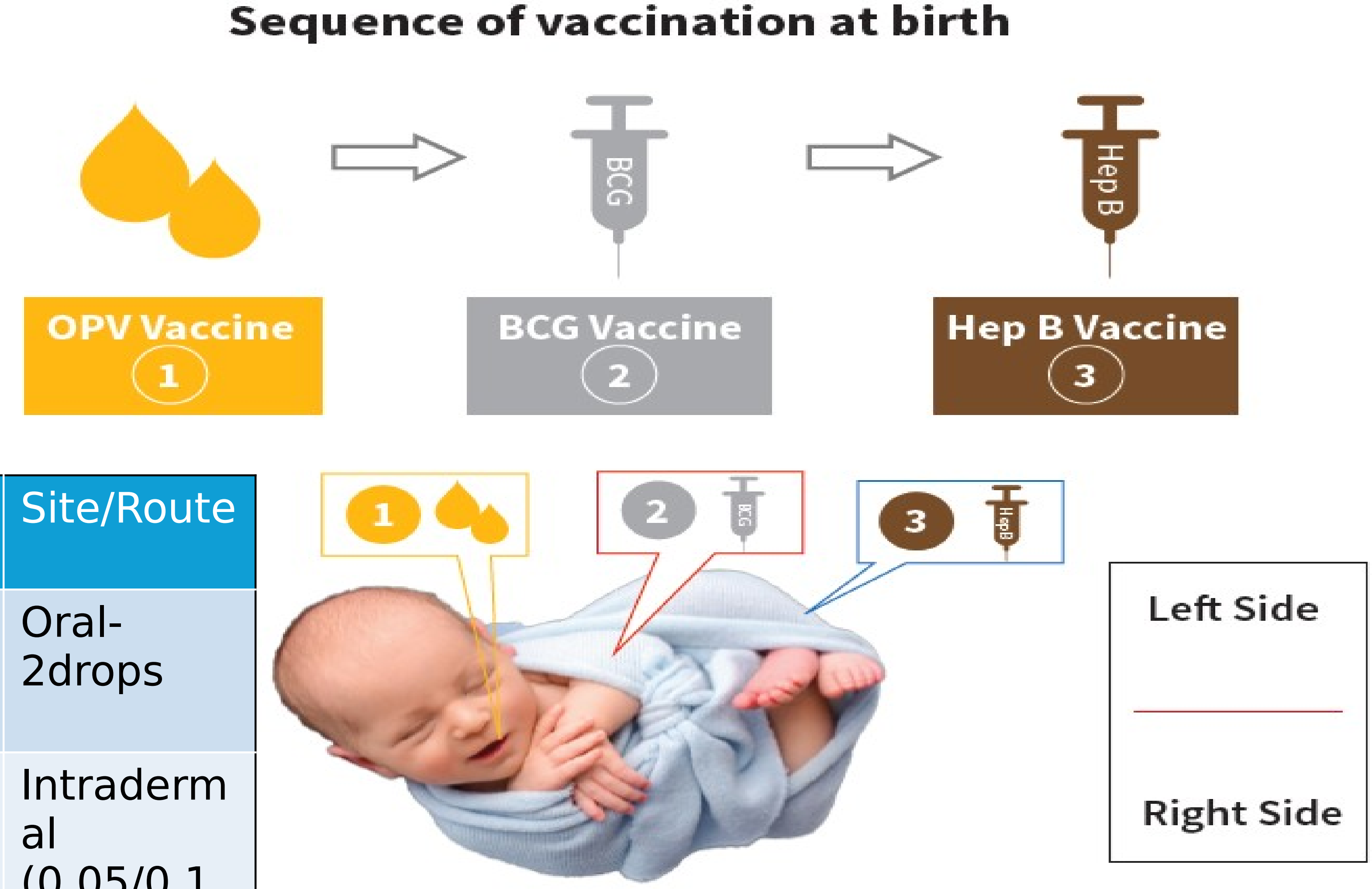
1. Poliomyelitis
2. Hepatitis B
3. Measles
4. Rubella
5. Diarrhoea due to Rota virus
6. Japanese Encephalitis

Bacterial diseases

1. Severe form of childhood Tuberculosis
2. Diphtheria
3. Pertussis
4. Tetanus
5. Haemophilus influenzae Type b related diseases
6. Pneumococcus related diseases

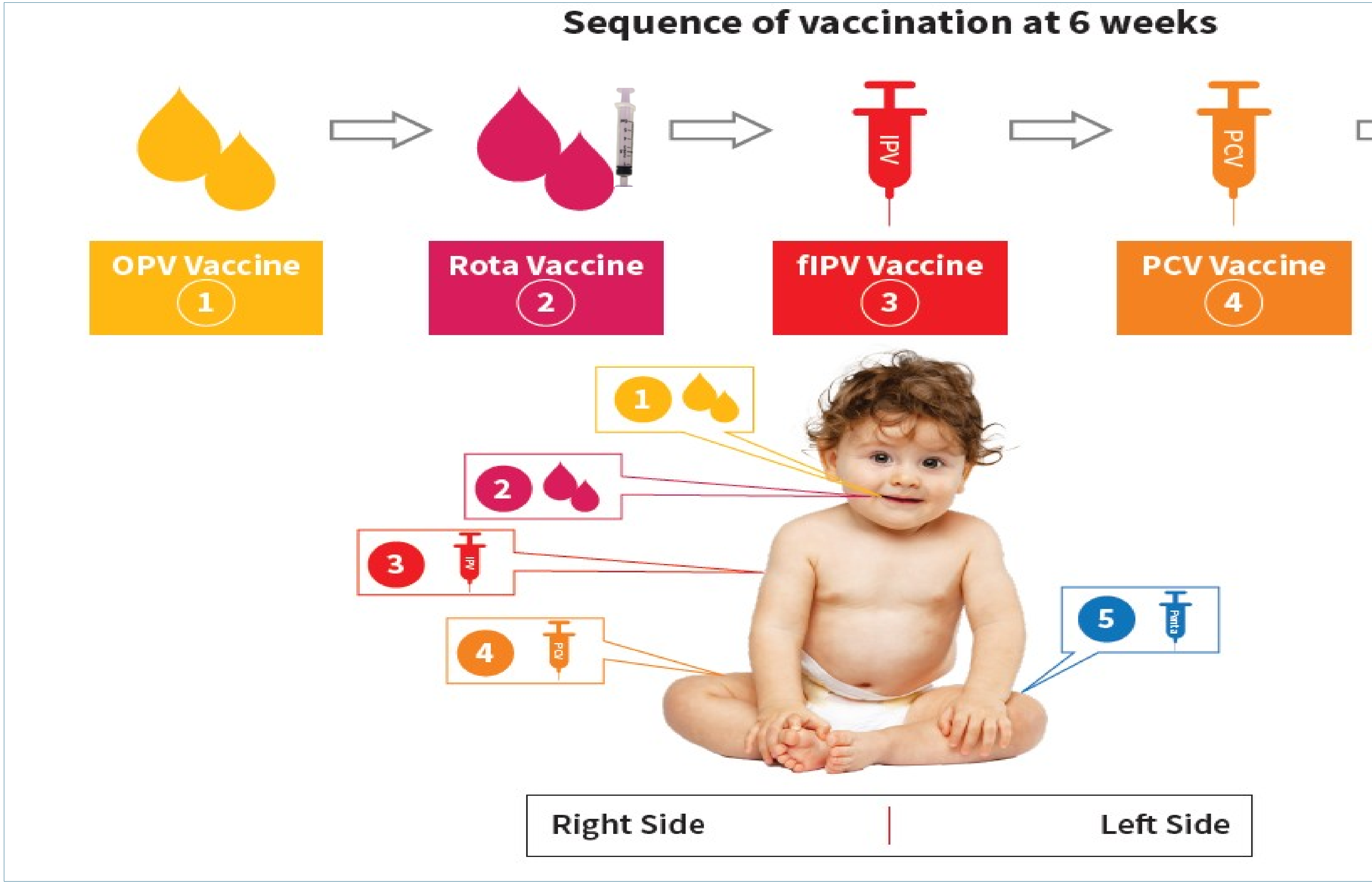
Revised Universal Immunization Schedule

Sequence , Site and Route of Vaccination at birth



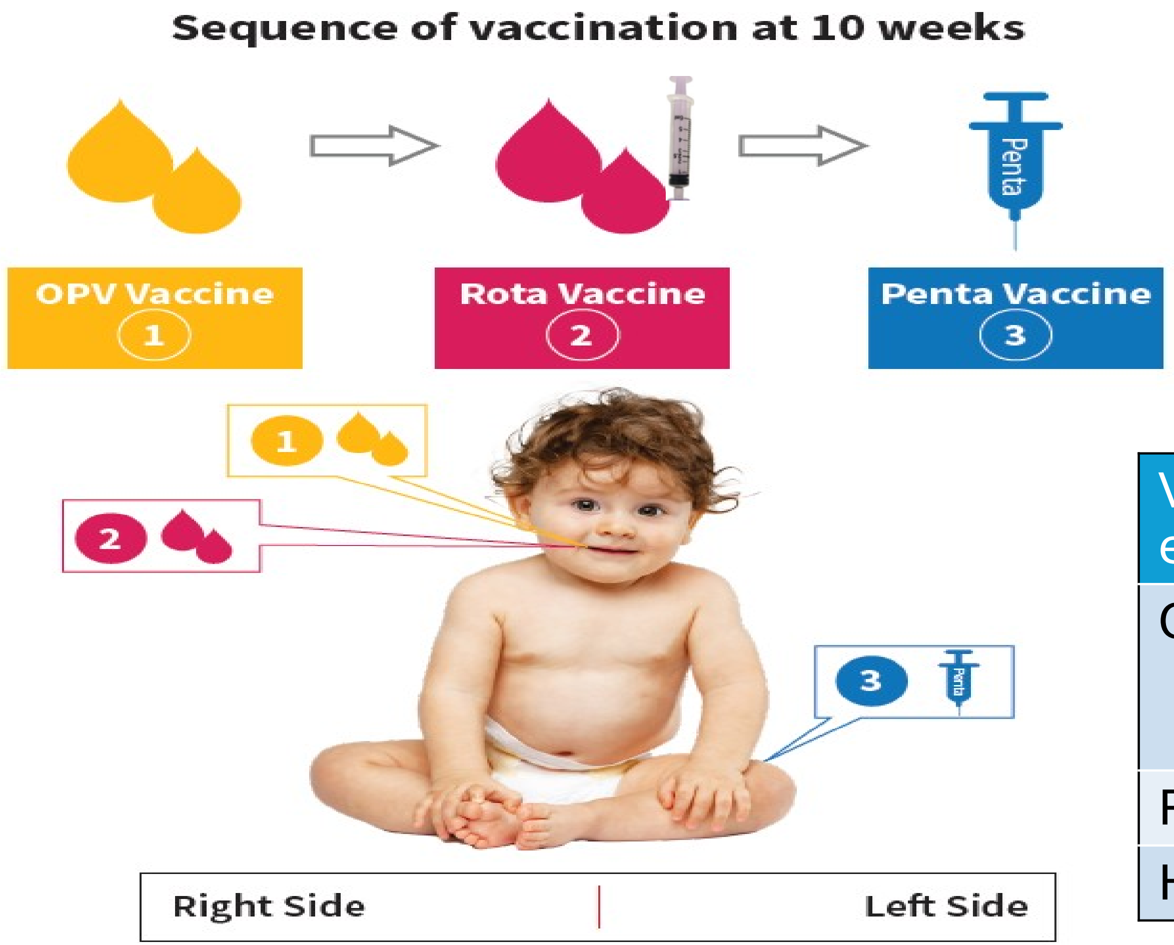
Vaccine	Site/Route
OPV	Oral-2drops
BCG	Intradermal (0.05/0.1 ml)

Sequence , Site and Route of Vaccination at 6th Week



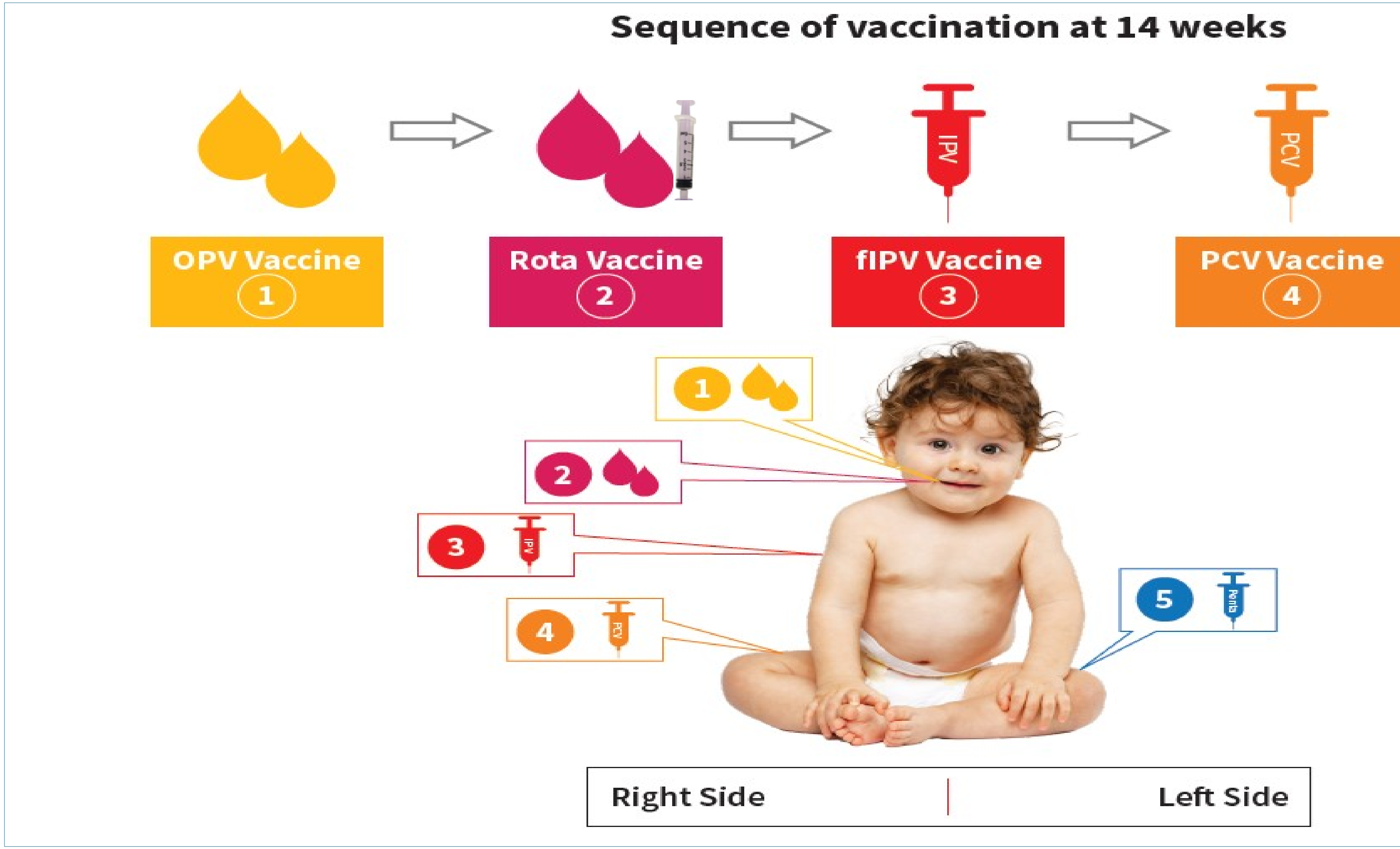
Vaccine	Site/Route
OPV	Oral-2drops
Rota	Oral-2ml
fIPV	Intradermal-0.1ml
PCV	IM-0.5ml
Hep-B	IM-0.5ml

Sequence , Site and Route of Vaccination at 10th Week



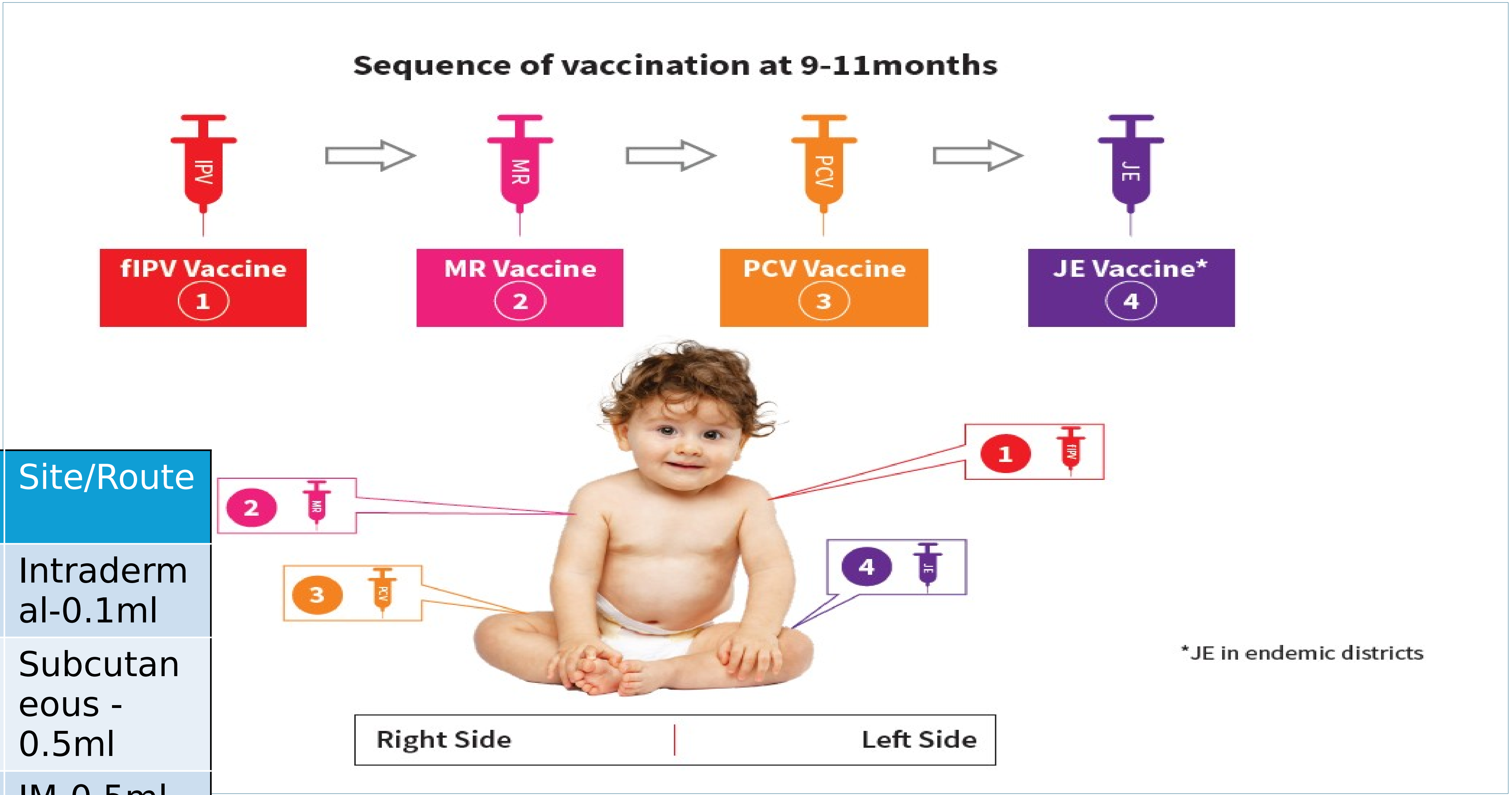
Vaccine	Site/Route
OPV	Oral-2drops
Rota	Oral-2ml
Hep-B	IM-0.5ml

Sequence , Site and Route of Vaccination at 14th Week



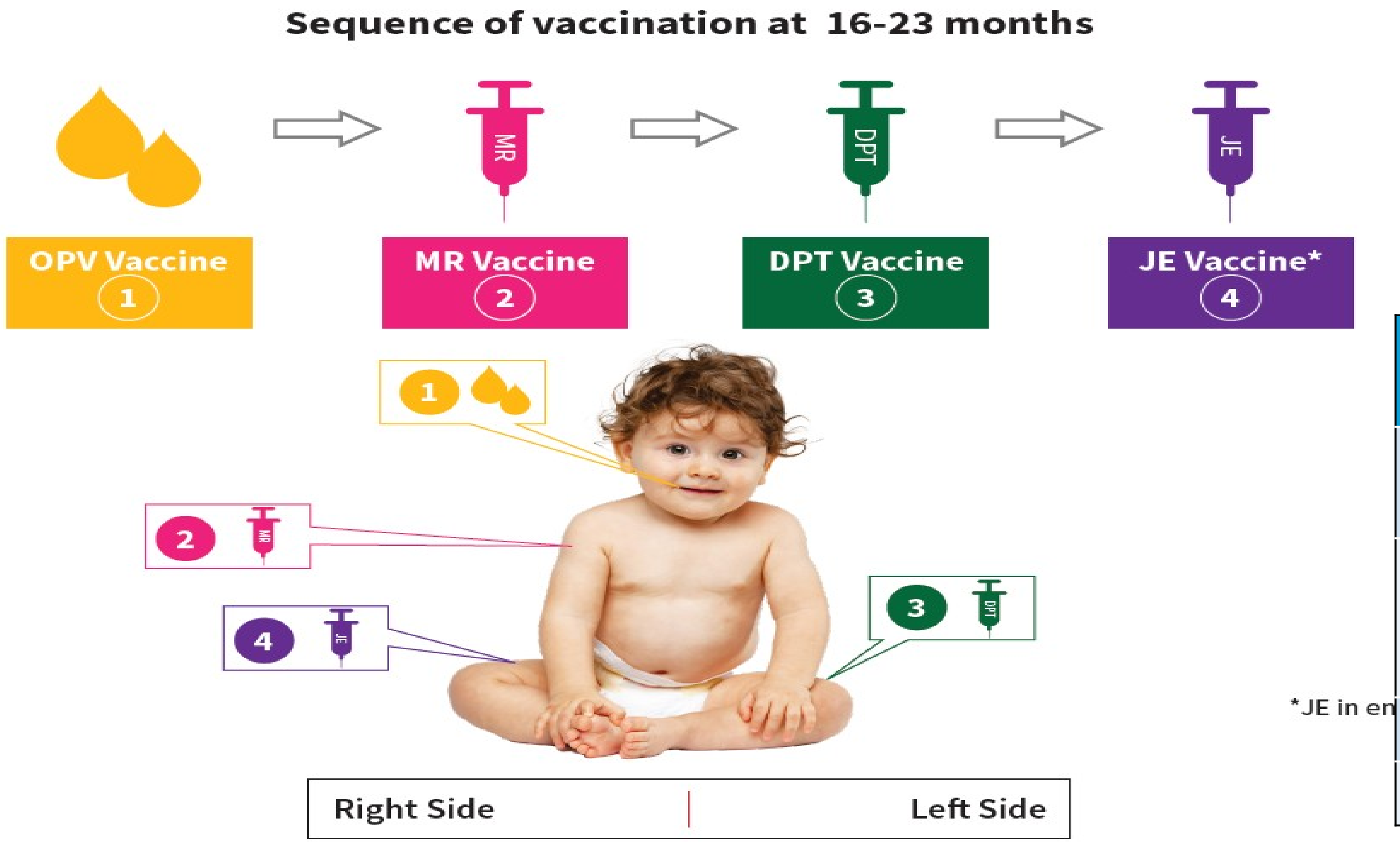
Vaccine	Site/Route
OPV	Oral-2drops
Rota	Oral-2ml
fIPV	Intradermal-0.1ml
PCV	IM-0.5ml
Penta	IM-0.5ml

Sequence , Site and Route of Vaccination at 9-11th month



Vaccine	Site/Route
fIPV	Intradermal-0.1ml
MR	Subcutaneous - 0.5ml
PCV	IM-0.5ml

Sequence , Site and Route of Vaccination at 16-23 months



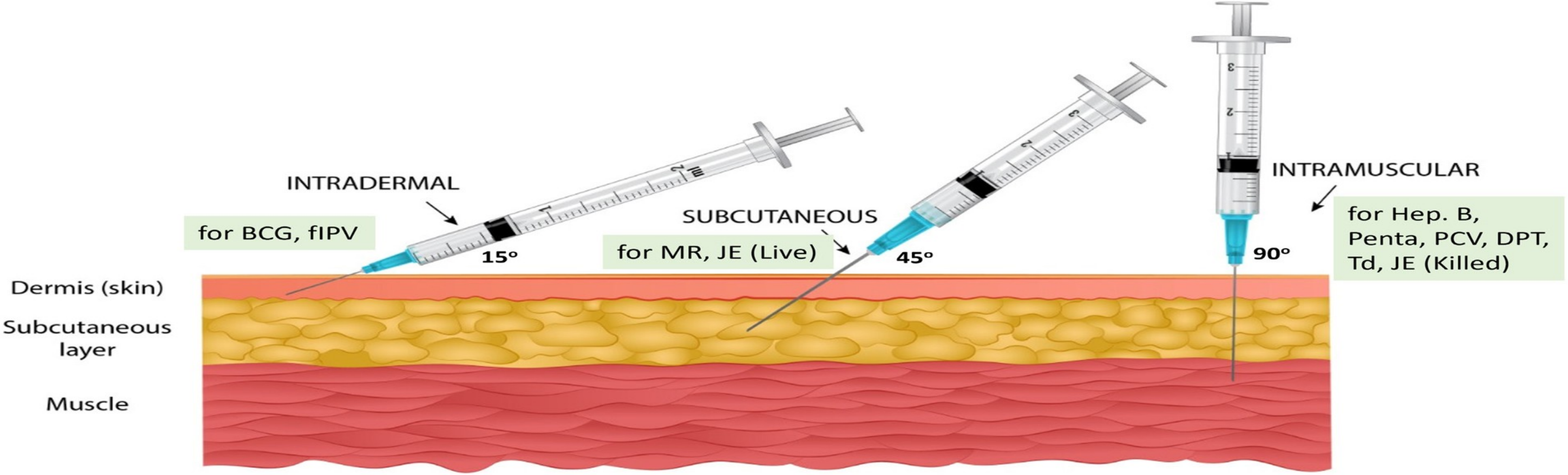
Vaccine	Site/Route
OPV	Oral-2drops
MR	Subcutaneous - 0.5ml
DPT	IM-0.5ml
JE	IM-0.5ml

Age	Vaccines given
5-6 years	DPT-Booster 2
10 years	Td
16 years	Td
Pregnant Mother	Td1, 2 or Td Booster**

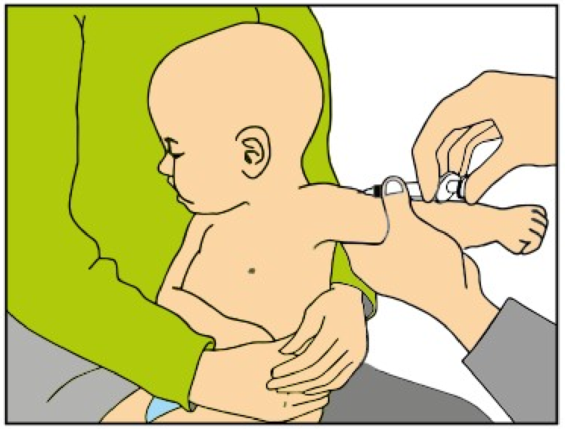
* *in endemic districts only*

** *one dose if previously vaccinated within 3 years*

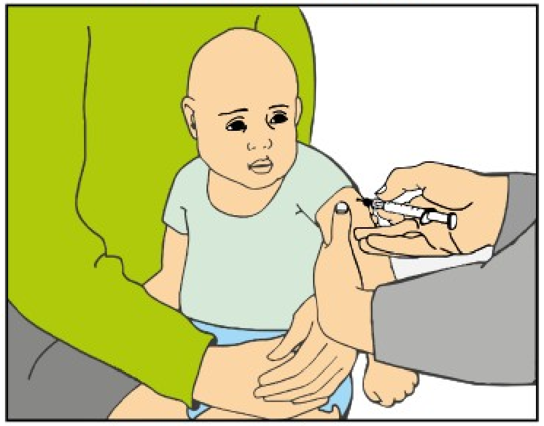
Syringe and needle position for Vaccination



Intradermal injection

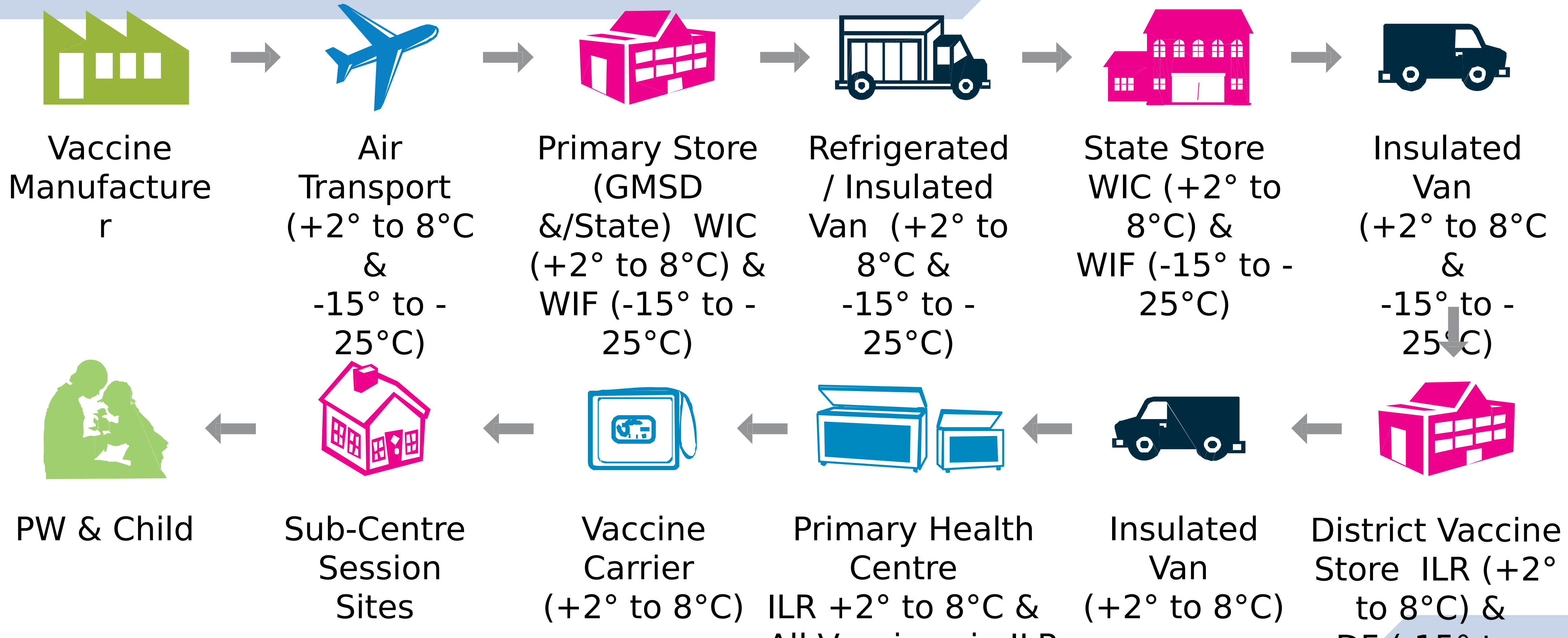


Subcutaneous injection



Intramuscular injection

Immunization cold-chain supply system



- **Ice lined refrigerator is used for storing the vaccine and Diluents**

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Sensitivity of Vaccine to heat, light and freezing

Vaccine	Exposure to heat/light	Exposure to cold
Heat & light sensitive vaccines		
OPV & Rota with VVM on label	Sensitive to heat	Not damaged by freezing
MR & BCG	Sensitive to heat & light	Not damaged by freezing
Freeze sensitive vaccines		
IPV, DPT, Pentavalent, JE & Rota with VVM on cap	Sensitive to heat	Damaged by freezing
Hepatitis B, PCV & Td	Relatively heat stable	Damaged by freezing

At the Sub district level, all vaccines are kept in the ILR at temperature of +2°C to +8°C

Vaccines sensitive to heat

Most

- OPV
- Rotavirus vaccine
- IPV
- DPT/Pentavalent/MR
- BCG/JE/Td
- PCV/HepB/ **HPV**

Least

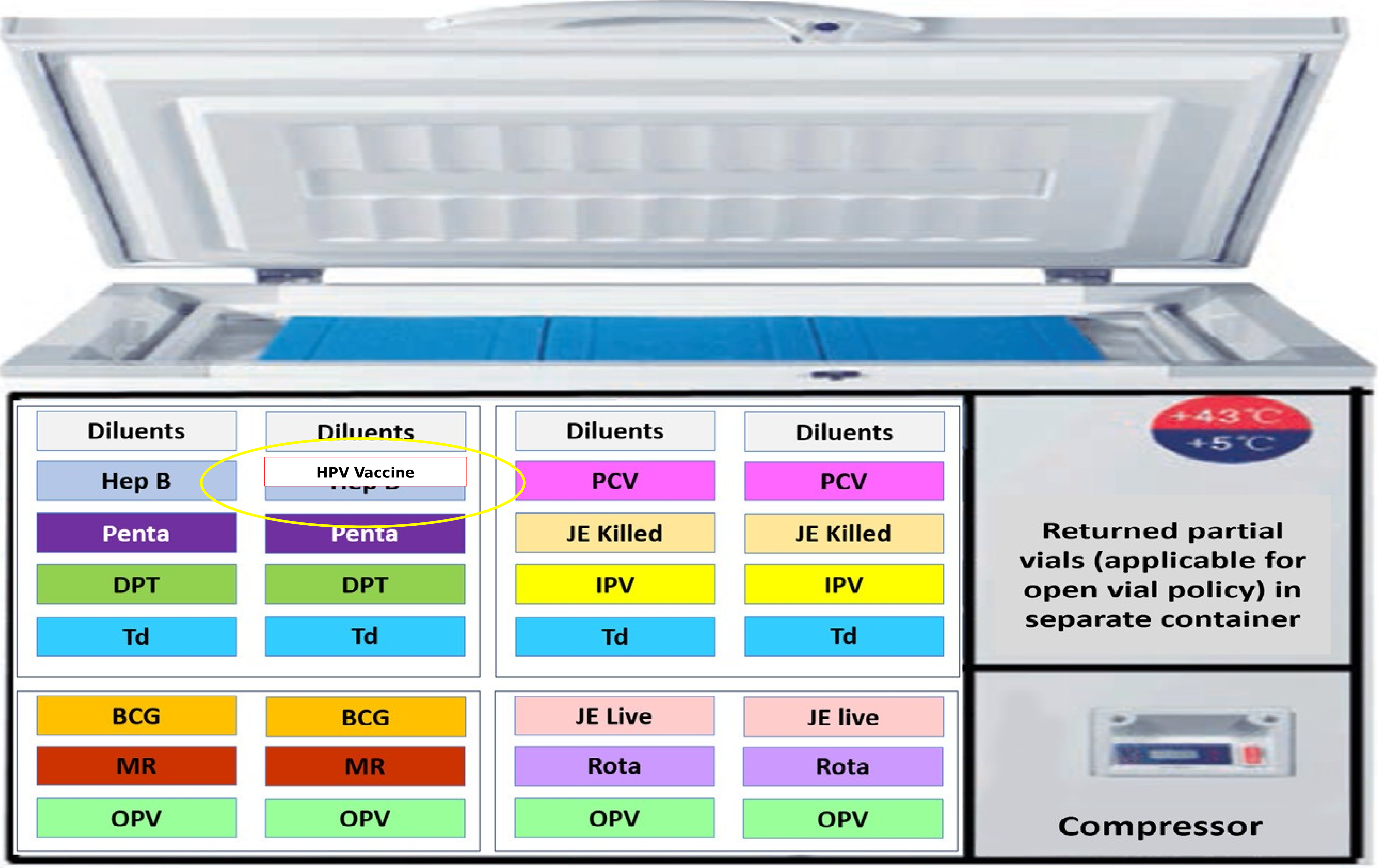
Vaccines sensitive to freezing

Most

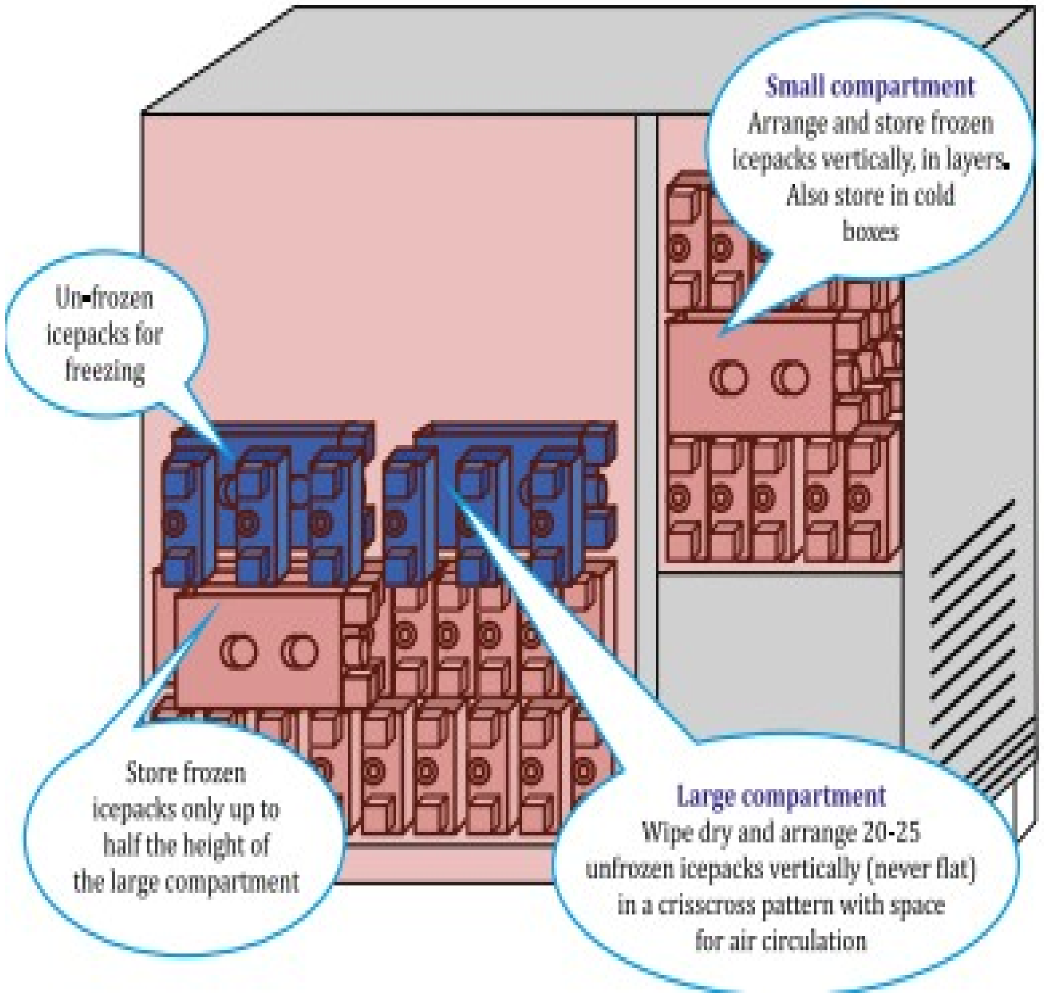
- DTP/Pentavalent/HepB/PCV/Td/ **HPV**
- IPV

Least

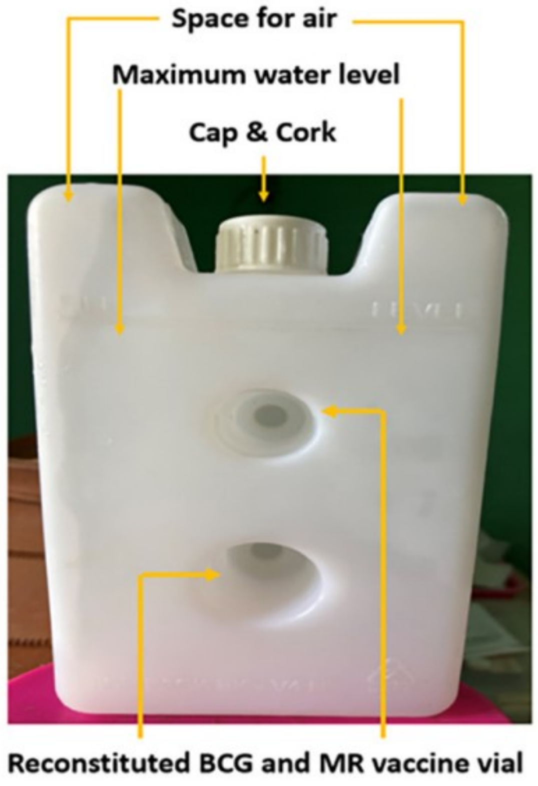
Vaccine Storage in ILR

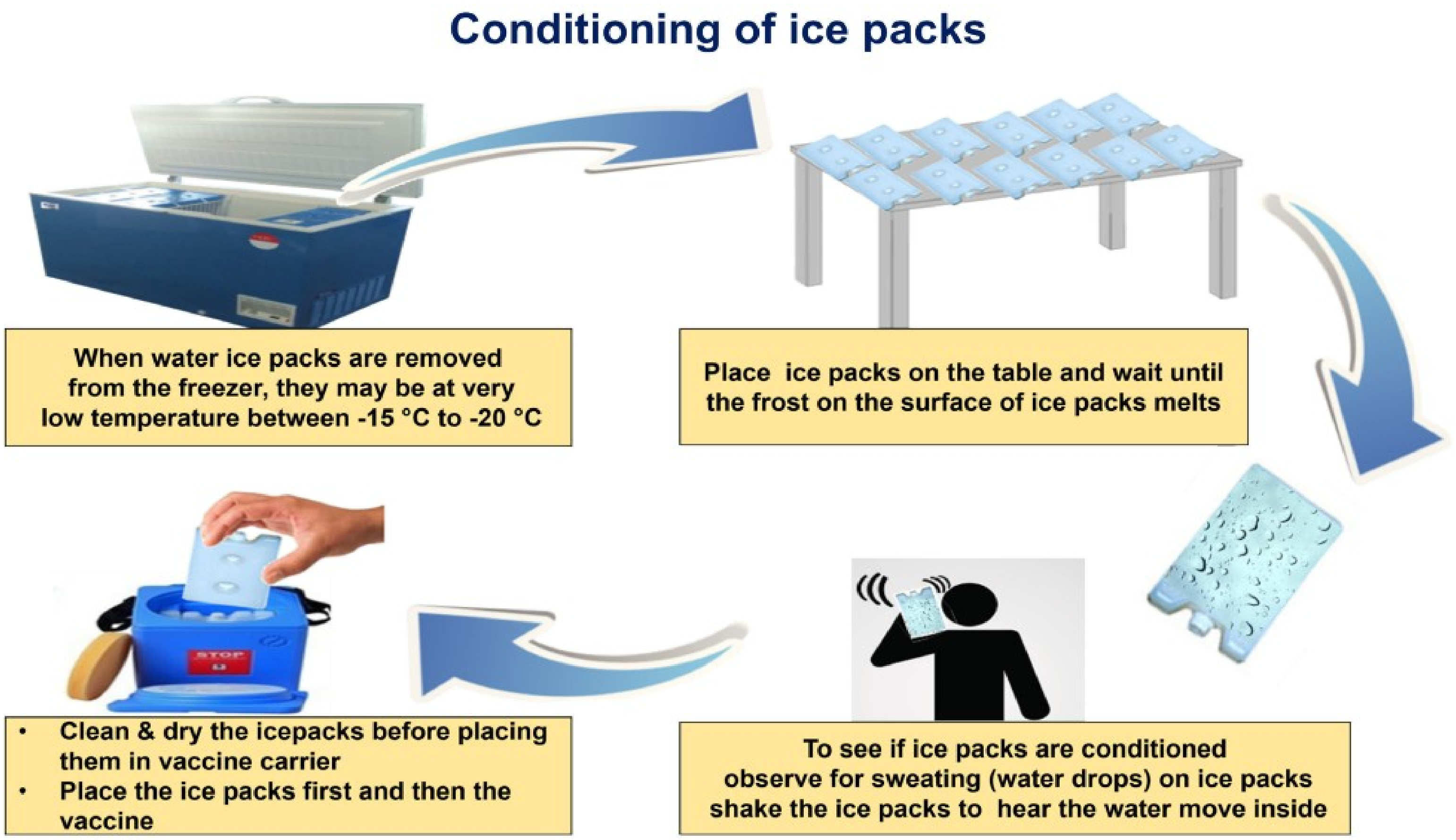


Storage of Icepacks in DF

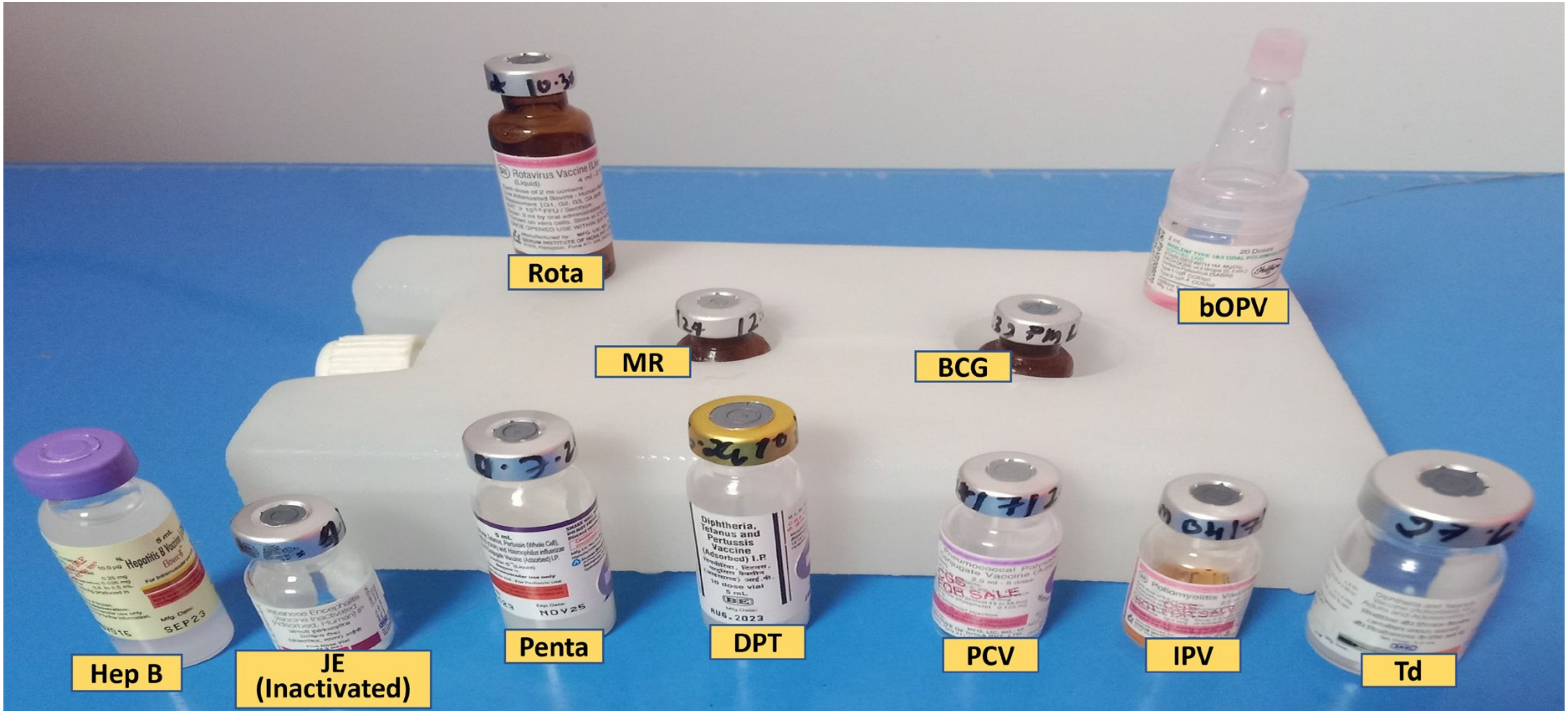


Conditioning of Ice Packs





Vaccine Arrangement at RI session sites



Adverse event following immunization (AEFI)

Definition : Adverse event following immunization (AEFI) is defined as any untoward medical occurrence which follows immunization, and which does not necessarily have a causal relationship with the usage of the vaccine.

Minor

Common, self-limiting reactions, usually occur within a few hours of injection and resolve after short period of time and pose little danger

Examples: pain, swelling at injection site, fever, irritability, malaise etc.

Severe

Can be disabling and rarely life threatening

do not lead to long-term problems

Examples of severe reactions include non-hospitalized cases of anaphylaxis that has recovered, high fever (>102-degree F), etc.

Serious

Death

Inpatient hospitalization

Persistent or significant disability

AEFI cluster

Congenital anomaly

Evokes significant parental / community concern

WHO cause specific definition of AEFIs

An AEFI that is caused or precipitated by a vaccine due to one or more of the inherent properties of the vaccine product.

Eg, Extensive limb swelling following DTP vaccination.

An AEFI that is caused or precipitated by a vaccine that is due to one or more quality defects of the vaccine product as provided by the manufacturer.

Eg, Failure by the manufacturer to completely inactivate a lot of inactivated polio vaccine leads to cases of paralytic polio.

An AEFI that is caused by Inappropriate vaccine handling, prescribing or administration.

Eg, Transmission of infection by contaminated multidose vial

An AEFI arising from anxiety about the immunization.

Eg, Vasovagal syncope in an adolescent following vaccination.

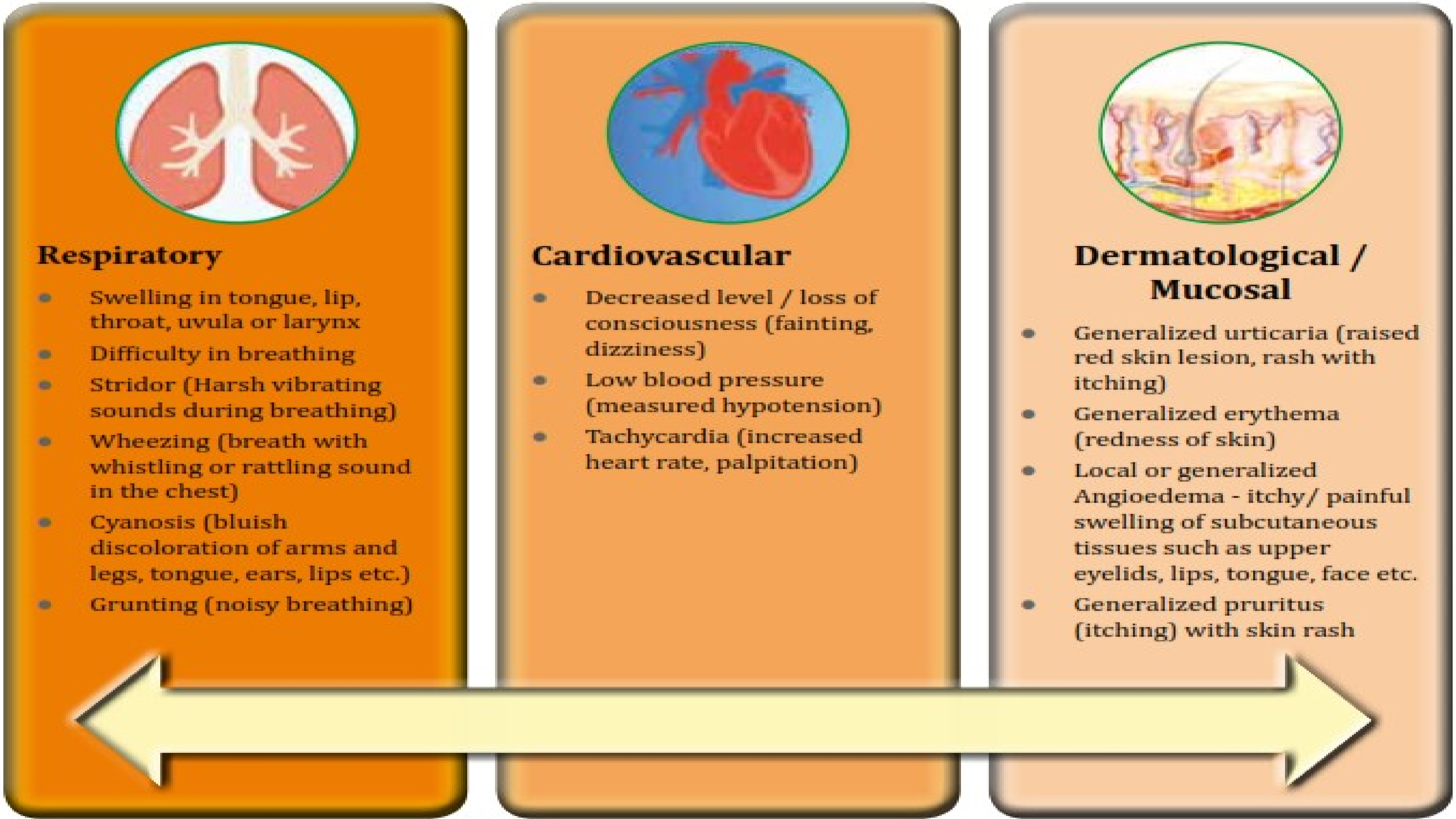
An AEFI that is caused by something other than the vaccine product, immunization error or immunization anxiety.

Eg, A fever occurs at the time of the vaccination, but is in fact caused by malaria.

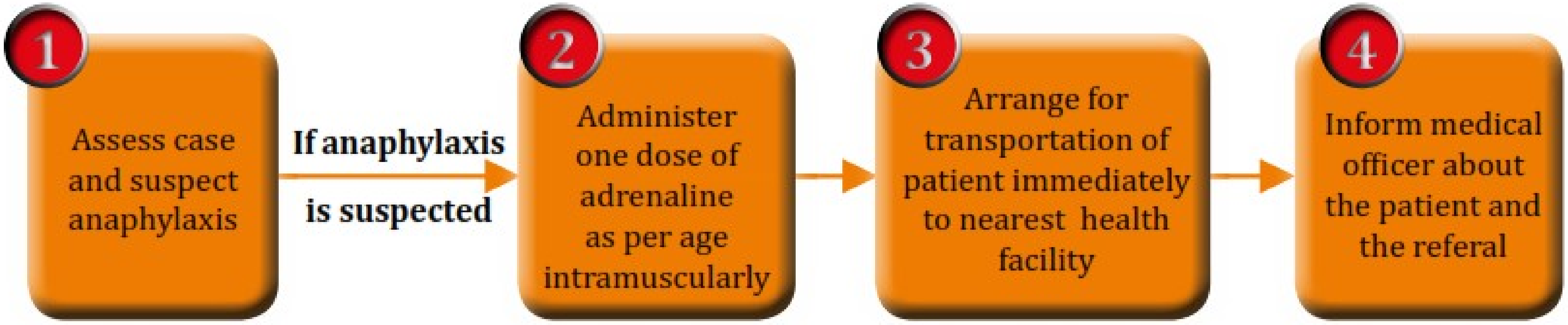
This is utilized for causality Assessment by State and National AEFI Committees

Recognition of Anaphylaxis

A case of anaphylaxis is suspected if following criteria is met: Early onset and rapid progression of more than one sign & symptoms of any of the two systems from the three systems – respiratory,



Steps for Initial Management of AEFI

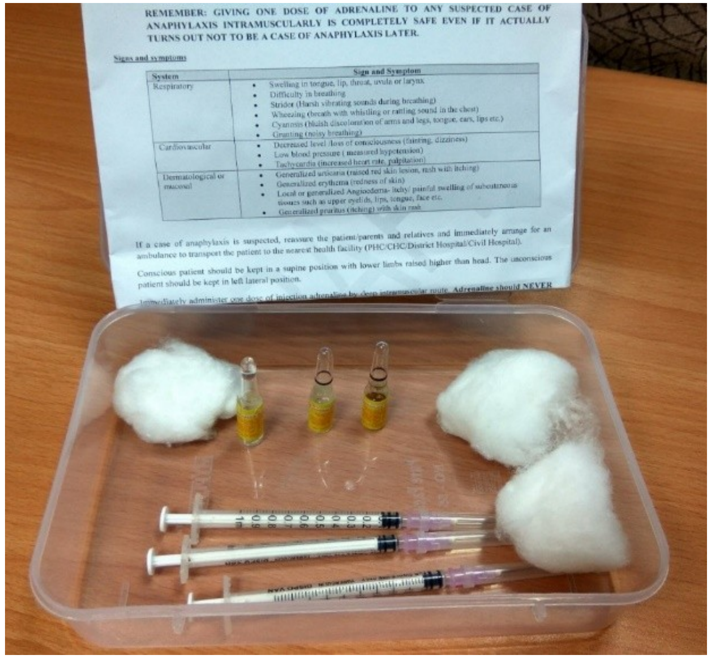


Managing AEFI

- **Ensure recipients wait for 30 minutes at session site after vaccination**
- After vaccination, inform the vaccine recipient
  - About any minor events which may occur
  - Visit the nearest health facility in case of any adverse event/discomfort/illness
- Anaphylaxis kit with inj. adrenaline within expiry date at outreach session site
- Inform Medical Officer immediately by telephone about serious/severe AEFIs
- Emergency numbers (102, 108, etc.) for transporting case to AEFI management center / higher health facility with vaccinator team

Contents of the Anaphylaxis Kit

1. Dose chart for adrenaline as per age
2. 1 mL ampoule of adrenaline (1:1000) – 3 nos.
3. 1 mL tuberculin syringes / 40 unit insulin **syringes with detachable needle** – 3 nos.
4. 24/25 G needles of 1 inch length – 3 nos.
5. Swabs – 3 nos.
6. Up to date contact information of Medical

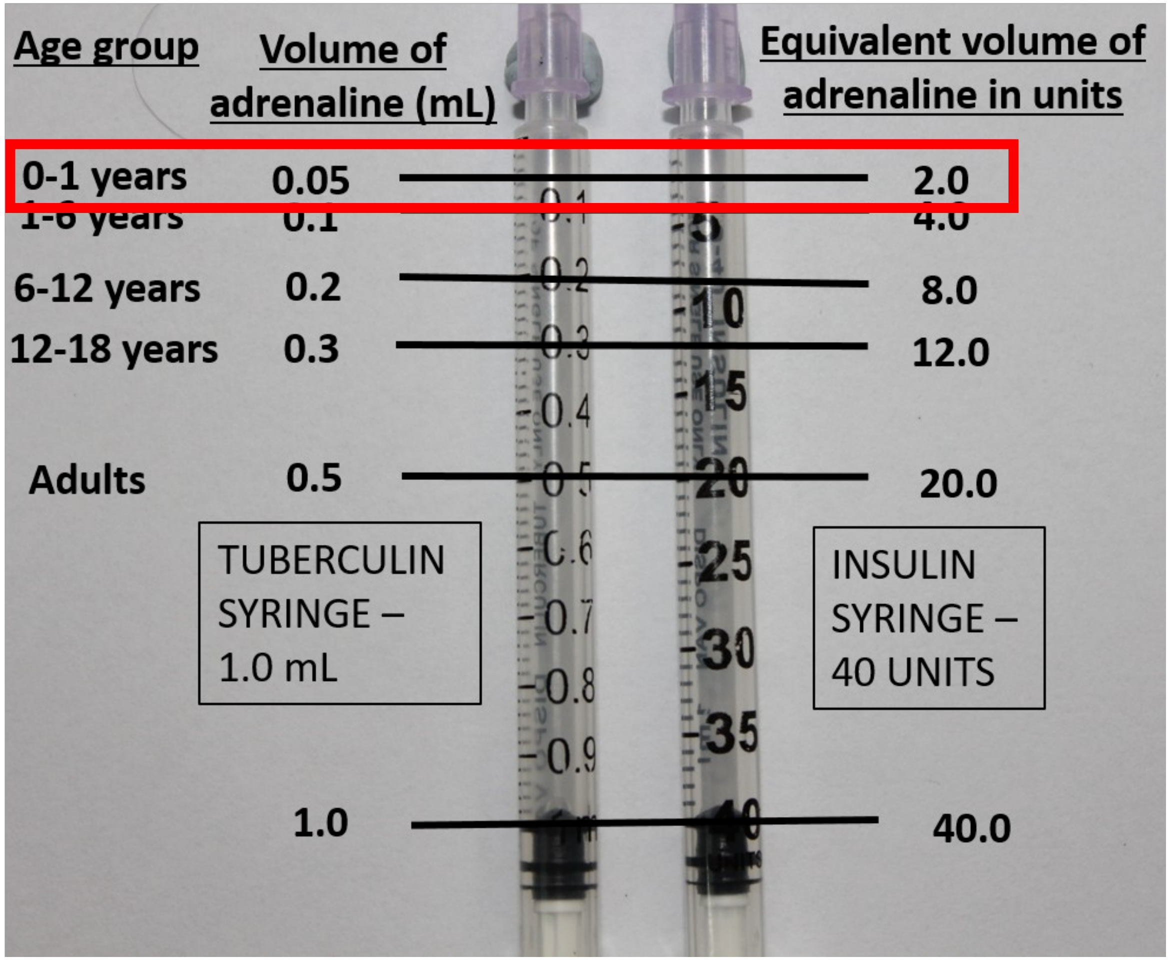


Anaphylaxis Kit

Dose of Adrenaline

- **Adrenaline is the drug of choice to treat anaphylaxis**
- **Ensure adequate stock of adrenaline for supplying to sessions**

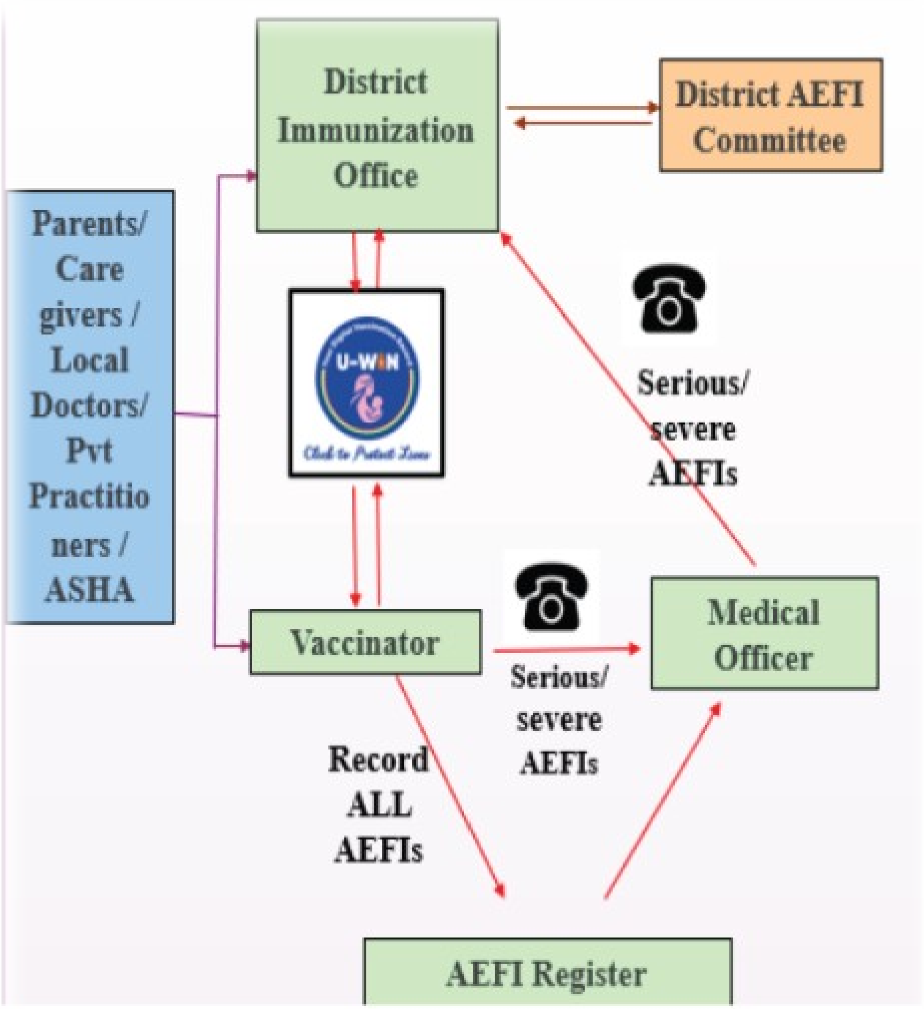
Age (in years)	Needle for intramuscular injection	Age specific dose of adrenaline (1:1000) in ml (tuberculin) or units (insulin)
Adults	1 inch 24G or 25G	0.5 ml / 20 units
12-18		0.3 ml / 12 units
6-12		0.2 ml / 8 units
1-6		0.1 ml / 4 units
0-1		0.05 ml / 2 units



Difference between AEFI Kit and Anaphylaxis kit

Difference between AEFI kit and Anaphylaxis kit

	AEFI kit	Anaphylaxis kit
Location	**At health facilities with Medical Officer**	**Outreach session**
For use by	**Medical Officer**	**ANM**
Contents		
Equipment for intubation and resuscitation	Yes	No
Ringer lactate, normal saline, 5% dextrose, IV drip set, scalp vein sets(2)	Yes	No
Inj. Hydrocortisone and Tab. Hydrocortisone	Yes	No
Cotton wool	Yes	Yes
Inj. Adrenaline ampoules	Yes	Yes
24G/25G needles 1 inch length	Yes	Yes
Tuberculin syringes (1ml) or Insulin syringes (40 units, without fixed needles)	Yes	Yes

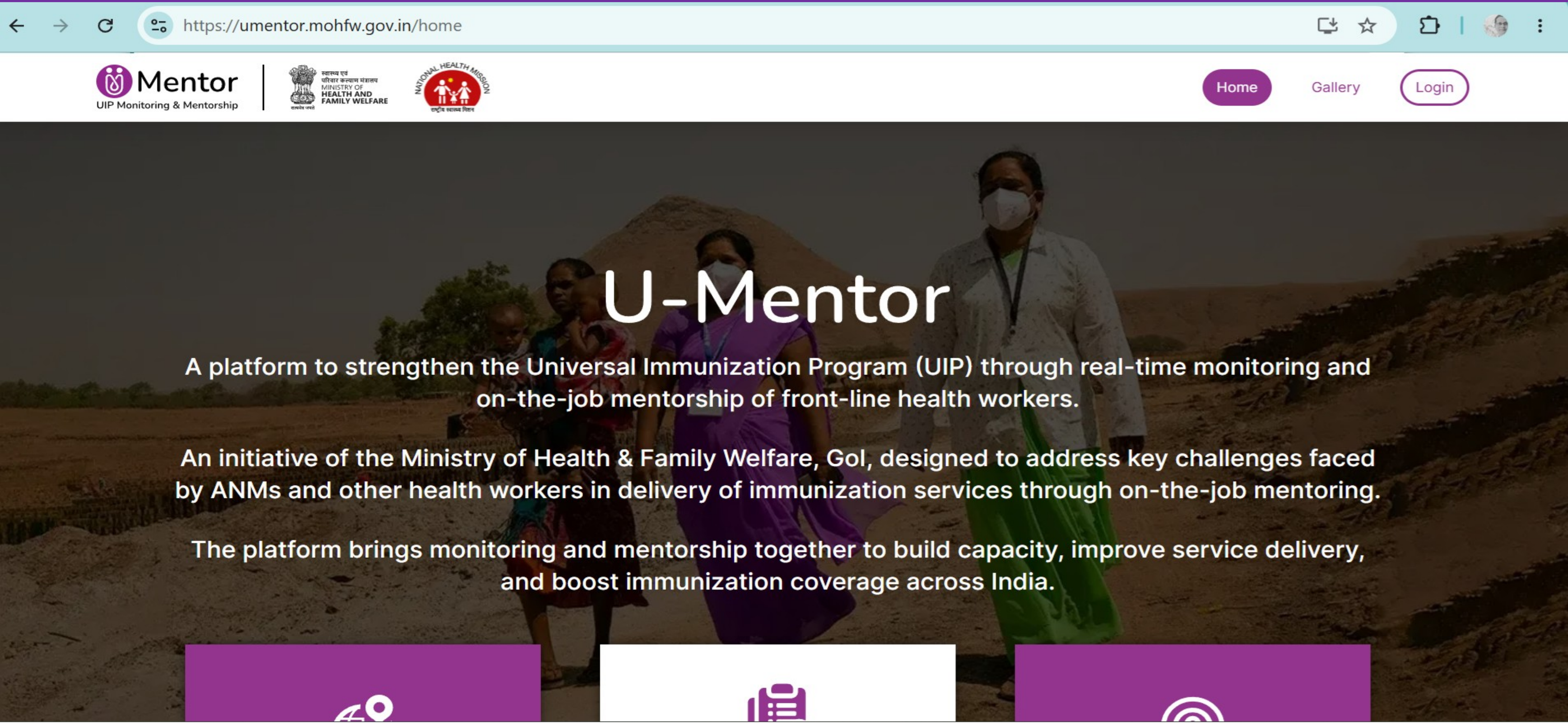


- All adverse events following any vaccination must be reported.
- An AEFI can be reported **anytime**, irrespective of the time duration between vaccination and adverse event or between adverse event and reporting.
- AEFI can be reported after **any vaccine**, whether:
  - Given in the UIP schedule or not
  - Given in routine or campaign mode
  - Given in a government or private setting
  - Beneficiary is a child or adult
- Record all AEFIs
  - **UIP vaccines-** minor, severe and serious ones through U-WIN- SAFEVAC and in AEFI registers (minor).
  - **Non-UIP vaccines-** severe & serious through SAFE-VAC.
  - AEFI registers should be available at health facilities like planning units and CCPs.

Immunization Wastes Segregation

Immunization Logistic	Particulars	Bags/Containers used
AD Syringes with packing	Cut hub of AD and disposable syringes	**White** [White coloured translucent, puncture-proof, leak-proof, tamper-proof containers]
	Plastic part of Syringe	**Red** [Red coloured non- chlorinated plastic bags (having thickness > 50 micron or containers)]
Vaccine parts	Vaccine adapter & dropper	
	Oral syringe	
Gloves	used gloves	
Ampoules	Ampoules (plastic)	
	Ampoules (glass)	**Blue** [Container/ cardboard boxes with blue coloured puncture-proof, temper-proof containers]
Vaccines	Expired vaccine or Discarded vaccine	
	Broken vials	
	Empty unbroken vials	
Cotton	Cotton contaminated with Blood or body fluid	**Yellow** [Yellow coloured non chlorinated plastic bags (having thickness > 50 micron or containers)]
Vitamins	Expired Bottles of Vitamins	
Paracetamol	Expired tab/ Syrup	

Monitoring and Supervision of RI Programme



Reporting and recording of RI Programme